Executive Summary: Key Findings and Recommendations

AnalystLab Africa Experience Lab | Week 3 | Track A: Data Analytics | Author: Divine Okezie Nwaoha | Tools: Power BI, Power Query, DAX, Python/SQL (validation)

Executive Summary

Problem. A healthcare organisation needs to understand encounter volumes, costs, payer coverage, procedure use and patient return visits. The data covers 27,891 encounters, 974 patients and 47,701 procedures (Jan 2011 to 5 Feb 2022, one hospital).

Approach. I profiled all six files, documented quality issues before cleaning anything, modelled encounters to patients, payers and procedures, built a 7-page Power BI dashboard, and independently checked key figures with SQL/Python.

Key findings

1. **Zero recorded payer coverage is the headline, and also the biggest trap.** 13,586 encounters (48.7%) show zero coverage, and they hold 62.2% of the $101.5M total claim cost. But five private insurers (Anthem, Cigna, Aetna, Humana, UnitedHealthcare) record about 0% coverage, while Medicaid, Medicare, Dual Eligible and BCBS record 69% to 94%. This points to a likely **data-capture gap**, not necessarily a funding gap.
2. **Urgent care costs far more than its volume suggests.** 13.1% of encounters but 23.0% of claim cost (average claim $6,369, against $2,894 for ambulatory).
3. **Ambulatory care drives volume** (44.9% of encounters), followed by outpatient (22.6%).
4. **Costs are concentrated in few patients.** The top 10% of patients (97 of 974) account for 70.3% of claim cost and 53.1% of encounters.
5. **Procedure cost rankings mislead without a volume floor.** "Admit to ICU" averages about $206K but occurred only 5 times. Frequent procedures are assessments, hospice care and dialysis.
6. **Only 0.27% of encounters (75) exceed 24 hours**, and some durations are implausible (up to about 5 years), which suggests timestamp errors.

Recommendations

#	Finding	Action	Benefit	Limitation / info needed
1	Five private insurers show about 0% coverage; zero-coverage encounters hold 62.2% of cost	Reconcile billing and remittance records for those insurers; fix coverage capture	A true view of funding gaps and revenue at risk	No remittance data supplied; cause unconfirmed
2	Urgent care: 13.1% of volume, 23.0% of cost	Review urgent care cost drivers and pricing/coding	Targets the largest cost-per-visit opportunity	No diagnosis or severity fields
3	10% of patients drive 70.3% of cost	Set up care-coordination review for the highest-cost patients	Focuses effort where cost concentrates	Cost is claim-level, not clinical need; no outcomes data
4	Rare high-cost procedures distort averages; 75 encounters have extreme durations	Budget frequent and rare procedures separately; add timestamp validation at entry	More reliable forecasts and duration KPIs	Base cost only; error source unknown

Limitations

Single hospital; 2011 is a ramp-up year and 2022 covers only 5 Feb; claim costs are not clinical outcomes; no diagnoses, so no clinical claims are made.